

Abhishek Sharma

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RESEARCH INTERESTS

Systems for large-scale AI/ML workloads — distributed and replicated systems (consensus, replication, fault tolerance, linearizability), machine learning systems (inference serving, scheduling, memory management), data-intensive and database systems, and performance/reliability engineering for high-performance and cloud computing pipelines.

EDUCATION

- **IOE, Pulchowk Campus (Tribhuvan University), Nepal** Nov. 2019 – Apr. 2024
Bachelor in Computer Engineering 76% average
- **SOSHGS, Gandaki, Pokhara, Nepal** 2017 – 2019
+2 Major in Physics and Mathematics 3.85 GPA (CGPA 3.72)

PUBLICATIONS

Topic Modeling for Nepali Political News — *IEEE ICICT 2025* | [Paper](#) | [Code](#)
Compared probabilistic (LDA), algebraic (NMF, LSA), and transformer-based (BERTopic) methods on a 2018–2023 Nepali political news corpus in a low-resource setting.

RESEARCH EXPERIENCE

- **Undergraduate Researcher** | *IOE Pulchowk Campus* | Supervisor: *Dr. Aman Shakya* 2024
 - Worked on a low-resource NLP project on Nepali political news using probabilistic, algebraic, and transformer-based methods. Published at IEEE ICICT 2025 (see Publications).
- **Product & Technical Design** | *ZeroTB* 2025 – 2026
 - Translated business requirements and stakeholder input into engineering flows, system architecture, and schema design for an AI-powered compliance-automation product, informed by competitor and USP analysis.

PROJECTS

- **Quasar: Linearizable Replicated Key-Value Store** | [Link](#) | *Raft, SMR, WAL, FastAPI, Python*
 - Implemented a Raft replicated state machine with term-based leader election, log-matching replication, majority-quorum commit, plus crash-durable WAL, snapshot compaction, and InstallSnapshot catch-up under crash and partition.
 - Served linearizable reads via a majority quorum check so an isolated leader cannot return stale state; built an interactive lab to step through election, partition, crash, and stale-leader recovery on a live three-node cluster.
- **Videolize: Text Script to Video Generator** | [Link](#) | *Stable Diffusion, Gemini API*
 - Script-to-video pipeline using Stable Diffusion, spherical latent interpolation, and Gemini API; 1st prize, GenAI category, Locus Hack-A-Week 2023.
- **Information Extraction from Invoices using LLM** | *GPT-4 Vision, YOLO, Table-Transformers*
 - Table and key-value extraction with GPT-4 Vision and fine-tuned Table-Transformers/YOLO; 1st prize, LLM category, Kathmandu University AI Competition 2023.
- **Procedural Terrain Generator** | [Link](#) | *C++, Perlin noise, Z-buffer, SFML*
 - Procedural heightmaps with Perlin noise in C++, rasterized from scratch (Bresenham lines, scanline fill, 2D clipping, Z-buffer) rather than a 3D engine.
- **University Network Infrastructure Design** | [Link](#) | *Cisco Packet Tracer*
 - Campus topology with subnetting, VLANs, DNS/web servers, and OSPF dynamic routing in Cisco Packet Tracer.

TEACHING EXPERIENCE

- **Instructor** | *LOCUS Fellowship, Pulchowk Campus* 2023
 - Co-designed curriculum and taught the mathematical foundations of machine learning (supervised learning, unsupervised learning, neural networks) to 100+ students, including implementations from first principles.
- **Programming Instructor** | *IT Club and PDSC, Pulchowk Campus* 2021 – 2022
 - Taught Django web development to 250+ students (PDSC) and Git/GitHub workflows to 40+ students in an in-person IT Club workshop.

PROFESSIONAL EXPERIENCE

Founding Software Engineer

ZeroTB

April 2025 – Present

San Francisco, CA / Kathmandu

- Designed and owned the backend architecture and APIs from the ground up for a compliance-automation product, including system-design and scaling decisions, taking it from zero to six production clients.
- Owned the production data layer: schema design, query performance, live-client migrations, and third-party integrations across the framework engine, trust center, and notification services.
- Built production RAG and LLM-orchestration pipelines (LangChain, LlamaIndex) for automated compliance document processing.

Associate Software Engineer (ML)

Docsumo

April 2024 – April 2025

Chabahil, Kathmandu

- Profiled the production key-value extraction pipeline and reduced end-to-end latency by $\sim 27\%$ (7–8 seconds), increasing document throughput.
- Owned the core document-extraction APIs for key-value and OMR, from research and implementation through optimization and refactor of the production serving path.
- Operated the Flask and MongoDB document-processing services on GCP, Kubernetes, Docker, and Redis for reliability and scale under high-volume traffic.

LEADERSHIP & MEMBERSHIP

President | *Society of Kaski Engineering Students (SOKES), Pulchowk Campus*

Apr 2023 – Feb 2024

- Led a student society of 200+ engineering students from Kaski District at Pulchowk Campus.

Secretary | *IT Club, Pulchowk Campus*

Jan 2022 – May 2023

- Department representative for Electronics and Computer Engineering; organized Git and design workshops and expanded outreach to students from other campuses.

AWARDS & ACHIEVEMENTS

Winner — LLM category in AI Competition organized by Kathmandu University [Dec. 2023] | [Link](#)

Winner — Generative AI category in LOCUS 2024 HackAWeek (Hackathon) [Jan. 2024]

Full Scholarship UnderGrad — **Rank 34** in IOE Examination (over 14000+ students) | [Link](#)

Entrance Topper — **Rank 1** in Kathmandu University [Electronics Department rank] (over 10000+ students) | [Link](#)

Golden Jubilee Scholarship — Awarded By Embassy of India in Kathmandu, Nepal

Entrance Topper — High School examination out of over 3000+ students

College Topper — High School (Grade 12) Board Examination [**3.85 GPA**]

School Topper — SEE Level Examination [**3.8 GPA**]

CERTIFICATIONS

Machine Learning Specialization — Coursera & DeepLearning.AI | [Link](#)

Deep Learning Specialization — Coursera & DeepLearning.AI | [Link](#)

Annual Nepal AI School — NAAMII,Nepal | [Link](#)

AWS Academy Graduate - AWS Academy Cloud Architecting — AWS | [Link](#)

AWS Academy Graduate - AWS Academy Data Engineering — AWS | [Link](#)

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, Dart, Haskell

Frameworks: FastAPI, Flask, Django, PyTorch, Node.js

Knowledge: Linux/Unix, Distributed Systems (Raft, SMR, linearizability), LLM systems (transformer inference, KV cache, RAG, agents), Computer Networks, Computer Vision (YOLO, CNN, ResNets, U-Net), Mathematical Domain (Linear Algebra, Probability and Statistics), NLP (RNNs, LSTMs, Transformers), DSA

Infra: Git, Docker, Kubernetes, Redis, PostgreSQL, MongoDB, GCP, AWS

LANGUAGE PROFICIENCY

English — IELTS 8.0 (Listening 8.5, Reading 8.5, Writing 7.0, Speaking 7.0)

Nepali — Native